# Manage FlexCache volumes

ONTAP 9.8 REST API reference

NetApp
February 13, 2024

This PDF was generated from https://docs.netapp.com/us-en/ontap-restapi-98/ontap/storage_flexcache_flexcaches_endpoint_overview.html on February 13, 2024. Always check docs.netapp.com for the latest.

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# Manage FlexCache volumes

## Storage flexcache flexcaches endpoint overview

### Overview

FlexCache is a persistent cache of an origin volume. An origin volume can only be a FlexVol while a FlexCache is always a FlexGroup.

The following relationship configurations are supported:

– Intra-Vserver where FlexCache and the corresponding origin volume reside in the same Vserver.

– Cross-Vserver but intra-cluster where FlexCache and the origin volume reside in the same cluster but belong to different Vservers.

– Cross-cluster where FlexCache and the origin volume reside in different clusters.

FlexCache supports fan-out and more than one FlexCache can be created from one origin volume. This API retrieves and manages FlexCache configurations in the cache cluster.

### FlexCache APIs

The following APIs can be used to perform operations related with FlexCache:

– GET /api/storage/flexcache/flexcaches

– GET /api/storage/flexcache/flexcaches/{uuid}

– POST /api/storage/flexcache/flexcaches

– DELETE /api/storage/flexcache/flexcaches/{uuid}

### Examples

#### Creating a FlexCache

The POST request is used to create a FlexCache.

```
# The API:
/api/storage/flexcache/flexcaches

# The call:
curl -X POST "https://<mgmt-ip>/api/storage/flexcache/flexcaches" -H
"accept: application/json" -H  "Content-Type: application/json" -d "{
\"aggregates\": [ { \"name\": \"aggr_1\" } ], \"name\": \"fc_333\",
\"origins\": [ {  \"svm\": { \"name\": \"vs_3\"  }, \"volume\": {
\"name\": \"vol_o1\" } } ], \"svm\": { \"name\": \"vs_1\" } }"

# The response:
{
"job": {
  "uuid": "e751dd5d-0f3c-11e9-8b2b-0050568e0b79",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/e751dd5d-0f3c-11e9-8b2b-0050568e0b79"
    }
  }
}
}
```

#### Retrieving FlexCache attributes

The GET request is used to retrieve FlexCache attributes. The object includes a large set of fields which can be expensive to retrieve. Most notably, the fields size, guarantee.type, aggregates, path, origins.ip_address, origins.size, and origins.state are expensive to retrieve. The recommended method to use this API is to filter and retrieve only the required fields.

```
# The API:
/api/storage/flexcache/flexcaches

# The call:
curl -X GET "https://<mgmt-ip>/api/storage/flexcache/flexcaches?" -H
"accept: application/json"

# The response:
{
  "records": [
    {
      "uuid": "04d5e07b-0ebe-11e9-8180-0050568e0b79",
      "name": "fc_322",
      "_links": {
        "self": {
          "href": "/api/storage/flexcache/flexcaches/04d5e07b-0ebe-11e9-
```

```
8180-0050568e0b79"
        }
      }
    },
    {
      "uuid": "47902654-0ea4-11e9-8180-0050568e0b79",
      "name": "fc_321",
      "_links": {
        "self": {
          "href": "/api/storage/flexcache/flexcaches/47902654-0ea4-11e9-
8180-0050568e0b79"
        }
      }
    },
    {
      "uuid": "77e911ff-0ebe-11e9-8180-0050568e0b79",
      "name": "fc_323",
      "_links": {
        "self": {
          "href": "/api/storage/flexcache/flexcaches/77e911ff-0ebe-11e9-
8180-0050568e0b79"
        }
      }
    },
    {
      "uuid": "ddb42bbc-0e95-11e9-8180-0050568e0b79",
      "name": "fc_32",
      "_links": {
        "self": {
          "href": "/api/storage/flexcache/flexcaches/ddb42bbc-0e95-11e9-
8180-0050568e0b79"
        }
      }
    },
    {
      "uuid": "ec774932-0f3c-11e9-8b2b-0050568e0b79",
      "name": "fc_333",
      "_links": {
        "self": {
          "href": "/api/storage/flexcache/flexcaches/ec774932-0f3c-11e9-
8b2b-0050568e0b79"
        }
      }
    }
  ],
  "num_records": 5,
```

```
  "_links": {
    "self": {
      "href": "/api/storage/flexcache/flexcaches?"
    }
  }
}
```

#### Retrieving the attributes of a FlexCache

The GET request is used to retrieve the attributes of a FlexCache. The object includes a large set of fields which can be expensive to retrieve. Most notably, the fields size, guarantee.type, aggregates, path, origins.ip_address, origins.size, and origins.state are expensive to retrieve. The recommended method to use this API is to filter and retrieve only the required fields.

```
# The API:
/api/storage/flexcache/flexcaches/{uuid}

# The call:
curl -X GET "https://<mgmt-ip>/api/storage/flexcache/flexcaches/ec774932-
0f3c-11e9-8b2b-0050568e0b79" -H  "accept: application/json"

# The response:
{
  "uuid": "ec774932-0f3c-11e9-8b2b-0050568e0b79",
  "name": "fc_333",
  "svm": {
    "name": "vs_1",
    "uuid": "e708fbe2-0e92-11e9-8180-0050568e0b79"
  },
  "size": 4294967296,
  "guarantee": {
      "type": "volume"
  },
  "aggregates": [
    {
      "name": "aggr_1",
      "uuid": "26f34b76-88f8-4a47-b5e0-d8e901fb1114"
    }
  ],
  "origins": [
    {
      "ip_address": "10.140.103.175",
      "size": 20971520,
      "create_time": "2019-01-03T15:19:55+05:30",
      "state": "online",
      "volume": {
```

```
        "name": "vol_o1",
        "uuid": "2bc957dd-2617-4afb-8d2f-66ac6070d313"
      },
      "svm": {
        "name": "vs_3",
        "uuid": "8aa2cd28-0e92-11e9-b391-0050568e4115"
      },
      "cluster": {
        "name": "node2",
        "uuid": "50733f81-0e90-11e9-b391-0050568e4115"
      }
    }
  ],
  "_links": {
    "self": {
      "href": "/api/storage/flexcache/flexcaches/ec774932-0f3c-11e9-8b2b-
0050568e0b79"
    }
  }
}
```

**Deleting a FlexCache**

The DELETE request is used to delete a FlexCache.

```
# The API:
/api/storage/flexcache/flexcaches

# The call:
curl -X DELETE "https://<mgmt-
ip>/api/storage/flexcache/flexcaches/ec774932-0f3c-11e9-8b2b-0050568e0b79"
-H  "accept: application/json"

# The response:
{
  "job": {
    "uuid": "e17994f2-0f3e-11e9-8b2b-0050568e0b79",
    "_links": {
      "self": {
        "href": "/api/cluster/jobs/e17994f2-0f3e-11e9-8b2b-0050568e0b79"
      }
    }
  }
}
```

# Retrieve a FlexCache volume in the cluster

GET `/storage/flexcache/flexcaches`

**Introduced In:** 9.6

Retrieves FlexCache in the cluster.

## Expensive properties

There is an added cost to retrieving values for these properties. They are not included by default in GET results and must be explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.

- `origins.ip_address` - IP address of origin.
- `origins.size` - Physical size of origin.
- `origins.state` - State of origin.
- `size` - Physical size of FlexCache.
- `guarantee.type` - Space guarantee style of FlexCache.
- `aggregates.name` or `aggregates.uuid` - Name or UUID of aggregrate of FlexCache volume.
- `path` - Fully-qualified path of the owning SVM's namespace where the FlexCache is mounted.

## Related ONTAP commands

- `volume flexcache show`

## Learn more

- DOC /storage/flexcache/flexcaches

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | query | False | Filter by uuid |
| aggregates.uuid | string | query | False | Filter by aggregates.uuid |
| aggregates.name | string | query | False | Filter by aggregates.name |
| constituents_per_aggregate | integer | query | False | Filter by constituents_per_aggregate |

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| origins.svm.uuid | string | query | False | Filter by origins.svm.uuid |
| origins.svm.name | string | query | False | Filter by origins.svm.name |
| origins.state | string | query | False | Filter by origins.state |
| origins.ip_address | string | query | False | Filter by origins.ip_address |
| origins.cluster.name | string | query | False | Filter by origins.cluster.name |
| origins.cluster.uuid | string | query | False | Filter by origins.cluster.uuid |
| origins.size | integer | query | False | Filter by origins.size |
| origins.volume.uuid | string | query | False | Filter by origins.volume.uuid |
| origins.volume.name | string | query | False | Filter by origins.volume.name |
| origins.create_time | string | query | False | Filter by origins.create_time |
| name | string | query | False | Filter by name |
| lock_consistent_when_disconnected | boolean | query | False | Filter by lock_consistent_when_disconnected<br>• Introduced in: 9.8 |
| size | integer | query | False | Filter by size |
| path | string | query | False | Filter by path |
| svm.uuid | string | query | False | Filter by svm.uuid |
| svm.name | string | query | False | Filter by svm.name |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| use_tiered_aggregate | boolean | query | False | Filter by use_tiered_aggregate<br>• Introduced in: 9.8 |
| guarantee.type | string | query | False | Filter by guarantee.type<br>• Introduced in: 9.7 |
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |
| desc] direction. Default direction is 'asc' for ascending. | return_records | boolean | query | False |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| num_records | integer | Number of records |
| records | array[flexcache] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "aggregates": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "aggr1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "guarantee": {
      "type": "volume"
    },
    "name": "vol1",
    "origins": {
      "cluster": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "cluster1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "create_time": "2018-06-04T19:00:00Z",
      "ip_address": "10.10.10.7",
      "size": 0,
      "state": "error",
      "svm": {
        "_links": {
```

```
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "svm1",
        "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
      },
      "volume": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "volume1",
        "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
      }
    },
    "path": "/user/my_fc",
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

aggregates

Aggregate

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

guarantee

| Name | Type | Description |
|---|---|---|
| type | string | The type of space guarantee of this volume in the aggregate. |

cluster

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

volume

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the volume. |
| uuid | string | Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move.<br>• example: 028baa66-41bd-11e9-81d5-00a0986138f7<br>• Introduced in: 9.6 |

flexcache_relationship

| Name | Type | Description |
| --- | --- | --- |
| cluster | cluster | |
| create_time | string | Creation time of the relationship. |
| ip_address | string | Cluster managerment IP of the remote cluster. |
| size | integer | Size of the remote volume. |
| state | string | Volume state |
| svm | svm | |
| volume | volume | |

svm

FlexCache SVM

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

flexcache

Defines the cache endpoint of FlexCache.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| aggregates | array[aggregates] | |
| constituents_per_aggregate | integer | Number of FlexCache constituents per aggregate when the 'aggregates' field is mentioned. |
| guarantee | guarantee | |
| lock_consistent_when_disconnected | boolean | Specifies if the origin will honor the cache side locks when doing the lock checks in the disconnected mode. The default value will be set to false to allow origin to skip such lock checks with the disconnected cache and go ahead with processing of the conflicting locks. If set to true the origin denies the lock operation (open or byte-range) if it conflicts with the authority granted to the disconnected cache. |
| name | string | FlexCache name |
| origins | array[flexcache_relationship] | |
| path | string | The fully-qualified path in the owning SVM's namespace at which the FlexCache is mounted. The path is case insensitive and must be unique within a SVM's namespace. Path must begin with '/' and must not end with '/'. Only one FlexCache be mounted at any given junction path. |

| Name | Type | Description |
|---|---|---|
| size | integer | Physical size of the FlexCache. The recommended size for a FlexCache is 10% of the origin volume. The minimum FlexCache constituent size is 1GB. |
| svm | svm | FlexCache SVM |
| use_tiered_aggregate | boolean | Specifies whether or not a Fabricpool-enabled aggregate can be used in FlexCache creation. The use_tiered_aggregate is only used when auto-provisioning a FlexCache volume. |
| uuid | string | FlexCache UUID. Unique identifier for the FlexCache. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Create a FlexCache volume in the cluster

POST `/storage/flexcache/flexcaches`

**Introduced In:** 9.6

Creates a FlexCache in the cluster.

## Required properties

- `name` - Name of FlexCache volume.
- `origins.volume.name` or `origins.volume.uuid` - Name or UUID of origin volume.
- `origins.svm.name` - Name of origin Vserver.
- `svm.name` or `svm.uuid` - Name or UUID of Vserver where FlexCache will be created.

## Recommended optional properties

- `path` - Path to mount the FlexCache volume

## Default property values

If not specified in POST, the following default property values are assigned:

- `size` - 10% of origin volume size or 1GB per constituent, whichever is greater.
- `guarantee.type` - Same as for a non-FlexCache FlexGroup volume.
- `constituents_per_aggregate` - 4 if aggregates.name or aggregates.uuid is used.
- `use_tiered_aggregate` - false if aggr-list is not used. This property is only used when auto-provisioning a FlexCache volume.
- `lock_consistent_when_disconnected` - false. This property specifies if the origin will honor the cache side locks when doing the lock checks in the disconnected mode.

## Related ONTAP commands

- `volume flexcache create`

## Learn more

- DOC /storage/flexcache/flexcaches

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| return_records | boolean | query | False | The default is false. If set to true, the records are returned.<br>• Default value: |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| aggregates | array[aggregates] | |

| Name | Type | Description |
| --- | --- | --- |
| constituents_per_aggregate | integer | Number of FlexCache constituents per aggregate when the 'aggregates' field is mentioned. |
| guarantee | guarantee | |
| lock_consistent_when_disconnected | boolean | Specifies if the origin will honor the cache side locks when doing the lock checks in the disconnected mode. The default value will be set to false to allow origin to skip such lock checks with the disconnected cache and go ahead with processing of the conflicting locks. If set to true the origin denies the lock operation (open or byte-range) if it conflicts with the authority granted to the disconnected cache. |
| name | string | FlexCache name |
| origins | array[flexcache_relationship] | |
| path | string | The fully-qualified path in the owning SVM's namespace at which the FlexCache is mounted. The path is case insensitive and must be unique within a SVM's namespace. Path must begin with '/' and must not end with '/'. Only one FlexCache be mounted at any given junction path. |
| size | integer | Physical size of the FlexCache. The recommended size for a FlexCache is 10% of the origin volume. The minimum FlexCache constituent size is 1GB. |
| svm | svm | FlexCache SVM |
| use_tiered_aggregate | boolean | Specifies whether or not a Fabricpool-enabled aggregate can be used in FlexCache creation. The use_tiered_aggregate is only used when auto-provisioning a FlexCache volume. |

| Name | Type | Description |
| --- | --- | --- |
| uuid | string | FlexCache UUID. Unique identifier for the FlexCache. |

**Example request**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "aggregates": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "aggr1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "guarantee": {
    "type": "volume"
  },
  "name": "vol1",
  "origins": {
    "cluster": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "cluster1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "create_time": "2018-06-04T19:00:00Z",
    "ip_address": "10.10.10.7",
    "size": 0,
    "state": "error",
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "volume": {
      "_links": {
```

```
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "volume1",
      "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
    }
  },
  "path": "/user/my_fc",
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```


## Response

```
Status: 202, Accepted
```

| Name | Type | Description |
|---|---|---|
| job | job_link | |

### Example response


```
{
  "job": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "string"
  }
}
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 66846870 | Either the SVM name or origin volume name is missing |
| 66846871 | Constituents per aggregate are specified but aggregate name is missing |
| 66846872 | More than one origin volume is specified |
| 66846873 | The specified SVM UUID is incorrect for the specified SVM name |
| 66846874 | The specified aggregate UUID is incorrect for the specified aggregate name |
| 66846875 | The specified aggregate name does not exist |
| 66846876 | The specified SVM does not exist or is not peered |
| 66846877 | The specified origin SVM name is of zero length |
| 66846878 | The specified SVM UUID is invalid |
| 66846730 | Failed to create a FlexCache volume |
| 66846760 | The specified SVM is not a data Vserver |
| 66846787 | The specified aggregate is a SnapLock aggregate |
| 66846812 | The specified aggregate is a Composite aggregate |
| 66846812 | The specified junction path is under a FlexCache volume |
| 66846834 | FlexCache encryption requires a cluster version of 9.6 or higher |
| 66846835 | A volume encryption license is not found |

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

aggregates

Aggregate

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

guarantee

| Name | Type | Description |
|---|---|---|
| type | string | The type of space guarantee of this volume in the aggregate. |

cluster

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

svm

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

volume

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the volume. |
| uuid | string | Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move.<br>• example: 028baa66-41bd-11e9-81d5-00a0986138f7<br>• Introduced in: 9.6 |

flexcache_relationship

| Name | Type | Description |
| --- | --- | --- |
| cluster | cluster | |
| create_time | string | Creation time of the relationship. |
| ip_address | string | Cluster managerment IP of the remote cluster. |
| size | integer | Size of the remote volume. |
| state | string | Volume state |
| svm | svm | |
| volume | volume | |

svm

FlexCache SVM

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

flexcache

Defines the cache endpoint of FlexCache.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| aggregates | array[aggregates] | |
| constituents_per_aggregate | integer | Number of FlexCache constituents per aggregate when the 'aggregates' field is mentioned. |
| guarantee | guarantee | |
| lock_consistent_when_disconnected | boolean | Specifies if the origin will honor the cache side locks when doing the lock checks in the disconnected mode. The default value will be set to false to allow origin to skip such lock checks with the disconnected cache and go ahead with processing of the conflicting locks. If set to true the origin denies the lock operation (open or byte-range) if it conflicts with the authority granted to the disconnected cache. |
| name | string | FlexCache name |
| origins | array[flexcache_relationship] | |
| path | string | The fully-qualified path in the owning SVM's namespace at which the FlexCache is mounted. The path is case insensitive and must be unique within a SVM's namespace. Path must begin with '/' and must not end with '/'. Only one FlexCache be mounted at any given junction path. |
| size | integer | Physical size of the FlexCache. The recommended size for a FlexCache is 10% of the origin volume. The minimum FlexCache constituent size is 1GB. |
| svm | svm | FlexCache SVM |

| Name | Type | Description |
|---|---|---|
| use_tiered_aggregate | boolean | Specifies whether or not a Fabricpool-enabled aggregate can be used in FlexCache creation. The use_tiered_aggregate is only used when auto-provisioning a FlexCache volume. |
| uuid | string | FlexCache UUID. Unique identifier for the FlexCache. |

job_link

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | The UUID of the asynchronous job that is triggered by a POST, PATCH, or DELETE operation. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Delete a FlexCache volume

DELETE `/storage/flexcache/flexcaches/{uuid}`

**Introduced In:** 9.6

Deletes a FlexCache. If a FlexCache volume is online, it is offlined before deletion.

## Related ONTAP commands

- `volume flexcache delete`

## Learn more

- DOC /storage/flexcache/flexcaches

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | Unique identifier of the FlexCache. |
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |

## Response

```
Status: 202, Accepted
```

| Name | Type | Description |
|---|---|---|
| job | job_link | |

### Example response

```
{
  "job": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "string"
  }
}
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 66846879 | The specified volume UUID is not a FlexCache volume |
| 66846731 | Failed to delete the FlexCache volume |
| 524546 | Failed to delete the FlexCache volume because the FlexCache volume is not unmounted |

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

job_link

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| uuid | string | The UUID of the asynchronous job that is triggered by a POST, PATCH, or DELETE operation. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Retrieve attributes of the FlexCache volume in the cluster

GET `/storage/flexcache/flexcaches/{uuid}`

**Introduced In:** 9.6

Retrieves attributes of the FlexCache in the cluster.

## Expensive properties

There is an added cost to retrieving values for these properties. They are included by default in GET. The recommended method to use this API is to filter and retrieve only the required fields. See Requesting specific fields to learn more.

- `origins.ip_address` - IP address of origin.
- `origins.size` - Physical size of origin.
- `origins.state` - State of origin.
- `size` - Physical size of FlexCache.
- `guarantee.type` - Space guarantee style of FlexCache.
- `aggregates.name` or `aggregates.uuid` - Name or UUID of aggregrate of FlexCache volume.
- `path` - Fully-qualified path of the owning SVM's namespace where the FlexCache is mounted.

## Related ONTAP commands

- `volume flexcache show`

## Learn more

- DOC /storage/flexcache/flexcaches

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | Unique identifier of FlexCache. |
| fields | array[string] | query | False | Specify the fields to return. |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| aggregates | array[aggregates] | |

| Name | Type | Description |
| --- | --- | --- |
| constituents_per_aggregate | integer | Number of FlexCache constituents per aggregate when the 'aggregates' field is mentioned. |
| guarantee | guarantee | |
| lock_consistent_when_disconnected | boolean | Specifies if the origin will honor the cache side locks when doing the lock checks in the disconnected mode. The default value will be set to false to allow origin to skip such lock checks with the disconnected cache and go ahead with processing of the conflicting locks. If set to true the origin denies the lock operation (open or byte-range) if it conflicts with the authority granted to the disconnected cache. |
| name | string | FlexCache name |
| origins | array[flexcache_relationship] | |
| path | string | The fully-qualified path in the owning SVM's namespace at which the FlexCache is mounted. The path is case insensitive and must be unique within a SVM's namespace. Path must begin with '/' and must not end with '/'. Only one FlexCache be mounted at any given junction path. |
| size | integer | Physical size of the FlexCache. The recommended size for a FlexCache is 10% of the origin volume. The minimum FlexCache constituent size is 1GB. |
| svm | svm | FlexCache SVM |
| use_tiered_aggregate | boolean | Specifies whether or not a Fabricpool-enabled aggregate can be used in FlexCache creation. The use_tiered_aggregate is only used when auto-provisioning a FlexCache volume. |

| Name | Type | Description |
| --- | --- | --- |
| uuid | string | FlexCache UUID. Unique identifier for the FlexCache. |

**Example response**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "aggregates": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "aggr1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "guarantee": {
    "type": "volume"
  },
  "name": "vol1",
  "origins": {
    "cluster": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "cluster1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "create_time": "2018-06-04T19:00:00Z",
    "ip_address": "10.10.10.7",
    "size": 0,
    "state": "error",
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "volume": {
      "_links": {
```

```
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "volume1",
      "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
    }
  },
  "path": "/user/my_fc",
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

### Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

aggregates

Aggregate

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

guarantee

| Name | Type | Description |
|---|---|---|
| type | string | The type of space guarantee of this volume in the aggregate. |

cluster

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

svm

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

volume

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the volume. |
| uuid | string | Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move.<br>• example: 028baa66-41bd-11e9-81d5-00a0986138f7<br>• Introduced in: 9.6 |

flexcache_relationship

| Name | Type | Description |
| --- | --- | --- |
| cluster | cluster | |
| create_time | string | Creation time of the relationship. |
| ip_address | string | Cluster managerment IP of the remote cluster. |
| size | integer | Size of the remote volume. |
| state | string | Volume state |
| svm | svm | |
| volume | volume | |

svm

FlexCache SVM

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

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